FATIMA KASSEM

Full Stack Web & Mobile Developer | Applied AI

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PROFESSIONAL SUMMARY

Full Stack Web and Mobile Developer working at a software company on a cross-platform Flutter product, now starting a Master 2 in Technology and Science of Information Systems after completing Master 1 in 2026. My My Master 1 research internship at Build4All was in applied AI: fine-tuning a multilingual open language model with QLoRA and building the intent-routing and retrieval layer around it. Experience across React, Next.js, Flutter, React Native, Java Spring Boot, Node.js and PostgreSQL, through employment, freelance client work and internships. Supervised around 10 interns across three internal company projects.

KEY SKILLS

AI / ML: Python, PyTorch, Hugging Face, LLM fine-tuning (QLoRA), mBERT, RAG / FAISS, TensorFlow, Keras, CNN

Frontend: React, Next.js, JavaScript, HTML5, CSS3, Tailwind CSS, Bootstrap, responsive design

Mobile: Flutter, React Native, Android (Java), cross-platform development

Backend: Java, Spring Boot, Node.js, REST APIs

Databases & Cloud: PostgreSQL, MySQL, SQLite, Firebase, cloud media storage

Tools: Git, GitHub, Postman, VS Code, IntelliJ IDEA, Android Studio, Figma

PROFESSIONAL EXPERIENCE

Full Stack Developer | 2026 - Present

Software Company (name confidential) - Lebanon

- Develop the company's multi-tenant cross-platform mobile product in Flutter, backed by Java Spring Boot and PostgreSQL.
- Supervised around 10 interns across three internal projects (Feb - May 2026): reviewed their code, guided their development process and helped resolve technical problems.

AI Research Intern - Master 1 | 2026

Build4All - Social Agent platform - Lebanon

- Benchmarked three small open language models and fine-tuned the selected 4B model with QLoRA on multilingual data (English, French, Arabic, Lebanese Arabizi).
- Proposed **Adaptive Guarded QLoRA**: a training method adding grounding, drift and repetition objectives whose weights are adjusted by a controller during training.
- Built the surrounding layer - an mBERT intent classifier, a router, retrieval over a controlled knowledge base and fact-grounded generation - evaluated with blinded scoring and bootstrap confidence intervals.
- In parallel, developed platform features full stack (Java Spring Boot, React): AI providers, image generation, Instagram and Facebook publishing, analytics, billing and trilingual RTL localisation.

Freelance Full Stack Developer | 2026

Client E-Commerce Project - Lebanon

- Delivered a production-ready e-commerce web application end to end as the sole developer, using Next.js and Java Spring Boot: catalog, categories, search and filters, admin dashboard and responsive design.

Full Stack Web Development Intern | May 2025 - Jul 2025

Laptop - NutriBite platform - Lebanon

- Independently designed and built NutriBite, a multi-role food e-commerce platform, with React, Node.js, PostgreSQL and Tailwind CSS.
- Delivered authentication, order and menu management and role-specific dashboards for customers, kitchens, delivery drivers and admins.

Full Stack Software Development Intern | Apr 2025 - Jun 2025

Hobby Sphere - supervised by Dr. Georges Lebbos - Lebanon

- Worked within a development team on a full-stack activity booking application, owning React Native screens and Spring Boot backend endpoints, and implemented the payment integration.

SELECTED PROJECTS

Multilingual AI Layer - Build4All Social Agent | *Master 1 Research, 2026* | *Python, PyTorch, QLoRA, mBERT, RAG*

- Self-hosted multilingual AI layer so the product no longer depends only on externally hosted commercial models, with intent routing, retrieval and grounded generation around a fine-tuned open model.

Wedding QR | *Multi-Tenant SaaS, 2026* | *Next.js, Java Spring Boot, PostgreSQL, Cloud Storage*

- Multi-tenant SaaS where each wedding gets its own QR code; guests scan it to open the event space and upload photos, stored in cloud storage rather than on the server.

NutriBite | *Solo Internship Project, 2025* | *React, Node.js, PostgreSQL* | *github.com/fatimahksm/nutribite*

- Multi-role food e-commerce platform for customers, kitchens, delivery drivers and admins, with order and menu management.

Hobby Sphere | *Team Project, 2025* | *React Native, Java Spring Boot, PostgreSQL* | *github.com/fatimahh0/HobbySphereMobile*

- Activity discovery and booking mobile application with profiles, reviews, notifications, maps and payment integration.

Brain MRI Tumor Classification | *University Team Project, 2026* | *Python, TensorFlow, Keras, CNN*

- Convolutional neural network trained to classify brain MRI scans for tumor detection, covering dataset preparation, preprocessing, training and evaluation.

EDUCATION

Master 2 - Technology and Science of Information Systems | Current

Lebanese University - Faculty of Technology - Lebanon

Master 1 - Technology and Science of Information Systems | Completed 2026

Lebanese University - Faculty of Technology - Lebanon

Bachelor's Degree - Business Computer (Informatique de Gestion) | 2022 - 2025

Lebanese University - Faculty of Technology - Lebanon

LANGUAGES

Arabic: Native | **French:** Good | **English:** Intermediate